

Pearson Edexcel Level 3 GCE

Friday 16 May 2025

Afternoon

**Paper
reference**

8FM0/27

Further Mathematics

Advanced Subsidiary

Further Mathematics options

27: Decision Mathematics 1

(Part of options D, F, H and K)

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
D1 Answer Book (enclosed)

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the answer book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.
- Do not return the question paper with the D1 Answer Book.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P74075A

©2025 Pearson Education Ltd.
Y:1/1/1/1/1/




Pearson

1

16 10 25 30 13 12 28 22 23 20

The list of ten numbers above is to be sorted into **ascending** order.

- (a) Carry out a bubble sort, starting at the left-hand end of the list, to produce the sorted list. You should only give the state of the list after each pass.

(3)

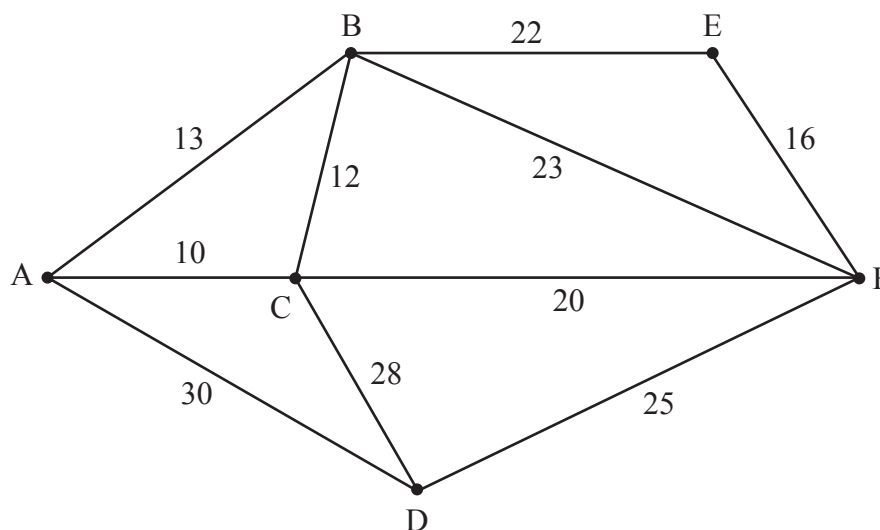


Figure 1

- (b) Use Prim's algorithm, starting at A, to find the minimum spanning tree for the network in Figure 1. You must list the arcs in the order in which you select them.

(2)

- (c) (i) Draw the minimum spanning tree on Diagram 1 in the answer book.

- (ii) Find the weight of the minimum spanning tree.

(2)

(Total for Question 1 is 7 marks)

Activity	Immediately preceding activities
A	-
B	-
C	-
D	B
E	A, B
F	B
G	B
H	C, D
I	G, H
J	E
K	E, F, I
L	G, H
M	G, H

- (a) Explain how you can deduce from the precedence table that at least one dummy will be needed when drawing the activity network. Your explanation should refer to specific activities. (1)
- (b) Draw the activity network described in the precedence table, using activity on arc. Your activity network must contain the minimum number of dummies only. (5)

Each activity in the precedence table takes 3 hours to complete.

- (c) State the minimum completion time. (1)

One of the activities now needs to be chosen to be extended by an hour. When the change is made the minimum completion time must not be affected.

- (d) List the activities that could be chosen. (1)

(Total for Question 2 is 8 marks)

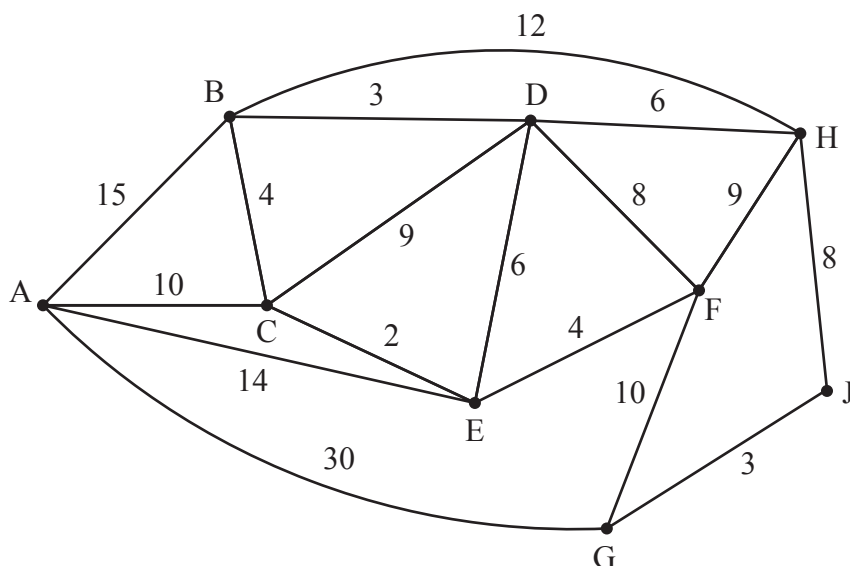


Figure 2

[The total weight of the network is 153]

Figure 2 models a network of roads in a town, where the nodes represent road junctions. The numbers on the edges are the times, in minutes, taken to walk along the corresponding roads.

- (a) State, with a reason, whether the graph in Figure 2 is Eulerian, semi-Eulerian or neither. (1)
- (b) (i) Use Dijkstra's algorithm to find the quickest path from A to J. (6)
- (ii) State the shortest time needed to walk from A to J.

Ruby manages road maintenance from her office located at junction B. She needs to walk along each road at least once, starting and finishing at her office. Road AE is temporarily blocked so she is unable to walk along it. Ruby wishes to minimise her journey time.

- (c) (i) Use an appropriate algorithm to find the roads that Ruby needs to traverse twice. You must make your method and working clear. (5)
- (ii) Calculate Ruby's journey time.

Ruby can save some time by choosing to start her route from any junction and to finish at her home at A. Road AE is still blocked and Ruby again wishes to minimise her journey time.

- (d) (i) Write down the junction at which Ruby should start. (2)
- (ii) Calculate how much time she would save.

(Total for Question 3 is 14 marks)

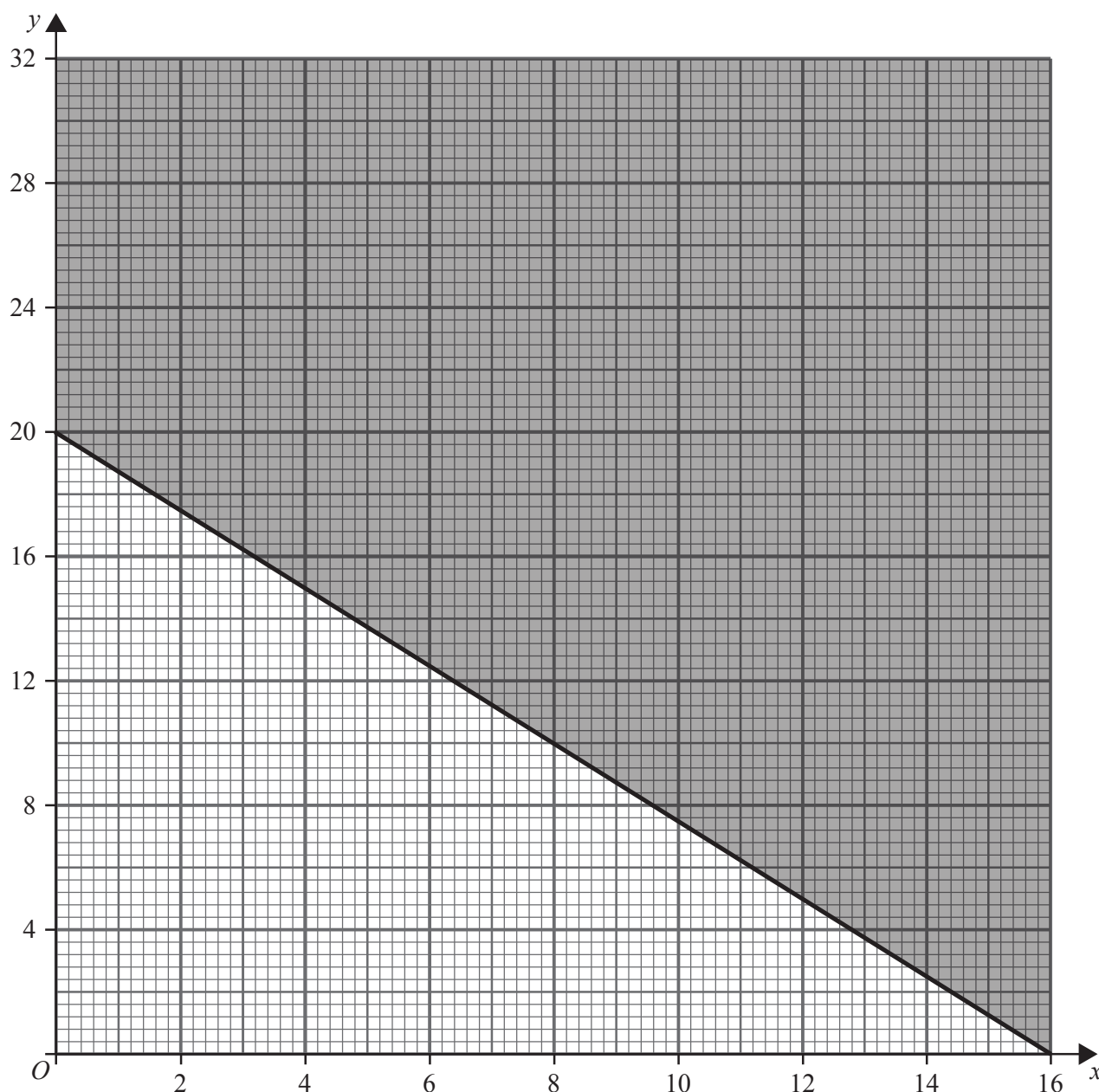


Figure 3

George is a baker who bakes x sponge cakes and y fruit cakes every day.

One of George's constraints is shown on Figure 3.

(a) Write down this constraint as a simplified inequality with integer coefficients.

(1)

Three further constraints are

- George must bake at least four fruit cakes every day
- for every two sponge cakes George bakes, he must bake at most three fruit cakes
- George needs 200 g of butter for each sponge cake and 100 g for each fruit cake and has 2.8 kg of butter available each day

- (b) (i) Write down the three inequalities which model these constraints. Give your answers in simplified form with integer coefficients.
- (ii) Add lines and shading to Diagram 1 in the answer book to represent these three constraints. Hence determine the feasible region and label it R .
- (5)

George makes £8 profit from each sponge cake he sells and £5 profit from each fruit cake he sells. Given that he wishes to maximise his profit,

- (c) use the vertex method to determine the number of sponge cakes and the number of fruit cakes George should bake each day. You must make your method and working clear.
- (5)

(Total for Question 4 is 11 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS



BLANK PAGE



BLANK PAGE



Pearson Edexcel Level 3 GCE

Friday 17 May 2024

Afternoon

Paper
reference

8FM0/27

Further Mathematics

Advanced Subsidiary

Further Mathematics options

27: Decision Mathematics 1

(Part of options D, F, H and K)

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
D1 Answer Book (enclosed)

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the answer book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.
- Do not return the question paper with the D1 Answer Book.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P75677A

©2024 Pearson Education Ltd.
F:1/1/1/




Pearson

1.

4 6.5 7 1.3 2 5 1.5 6 4.5 6 1

The list of eleven numbers shown above is to be sorted into **descending** order.

- (a) Carry out a **quick sort** to produce the sorted list. You should show the result of each pass and identify the pivots clearly. (3)
- (b) Use the **first-fit decreasing** bin packing algorithm to pack the numbers into bins of size 10 (3)
- (c) Determine whether your answer to part (b) uses the minimum number of bins. You must justify your answer. (2)

A **different** list of eleven numbers is to be sorted into **descending** order using a **bubble sort**. The list after the **second** pass is

4.5 5.6 3.8 6.7 5.4 1.6 4.8 9.1 3.3 1.7 1.5

- (d) Explain how you know that at least one of the first two passes of the bubble sort was **not** carried out correctly. (1)

(Total for Question 1 is 9 marks)

2. A company manages an awards evening.

The table below lists the activities required to set up the room for the evening, and their immediately preceding activities. Each activity requires exactly one person.

Activity	Immediately preceding activities
A	-
B	A
C	A
D	C
E	C
F	B, D, E
G	E
H	B
J	H, F, G

Figure 1 shows a partially completed activity network used to model the project. Each activity is represented by an arc.

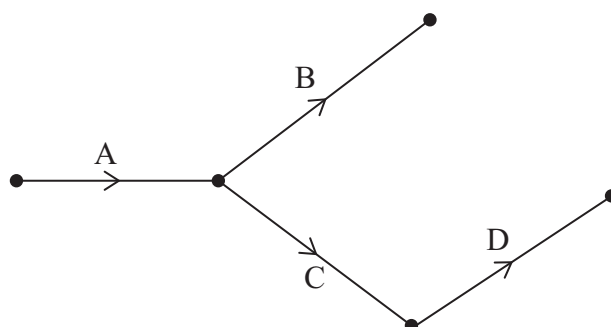


Figure 1

- (a) Add the remaining five activities to Diagram 1 in the answer book to complete the activity network, using exactly two dummies.

(3)

In addition to setting up the room, the company must prepare the meals for the guests. Figure 2 shows the activity network for preparing the main courses. The numbers in brackets represent the time, in minutes, to complete each task.

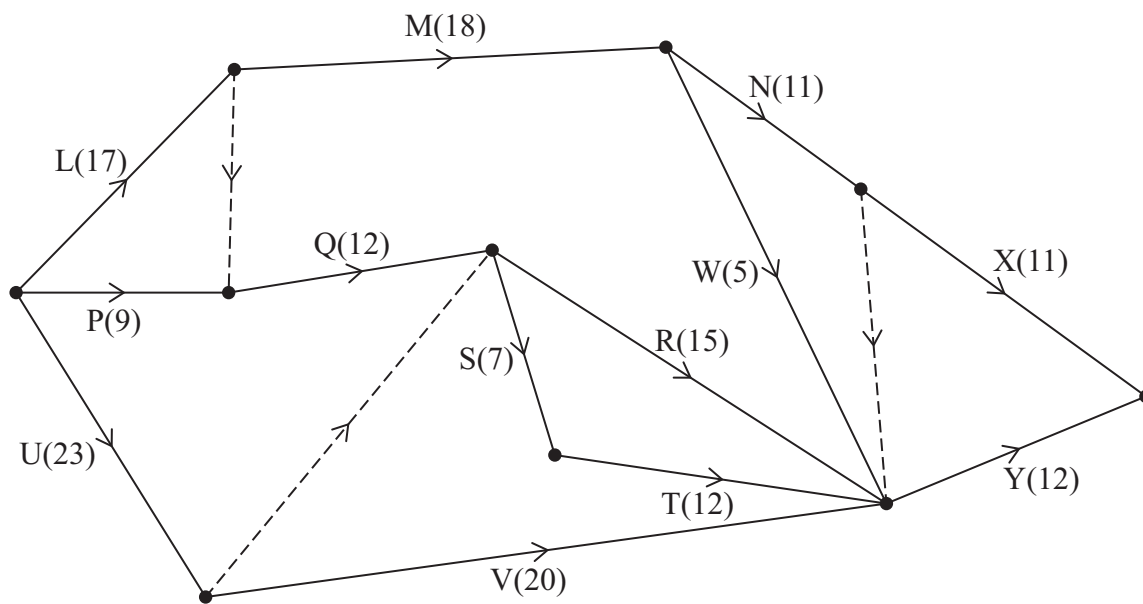


Figure 2

- (b) Complete Diagram 2 in the answer book to show the early event times and the late event times for the activity network shown in Figure 2.

(3)

- (c) State the critical activities.

(1)

- (d) Given that the main courses need to be ready to be served (with all activities completed) at 8 pm, state the latest time that activity R can start.

(1)

(Total for Question 2 is 8 marks)

3.

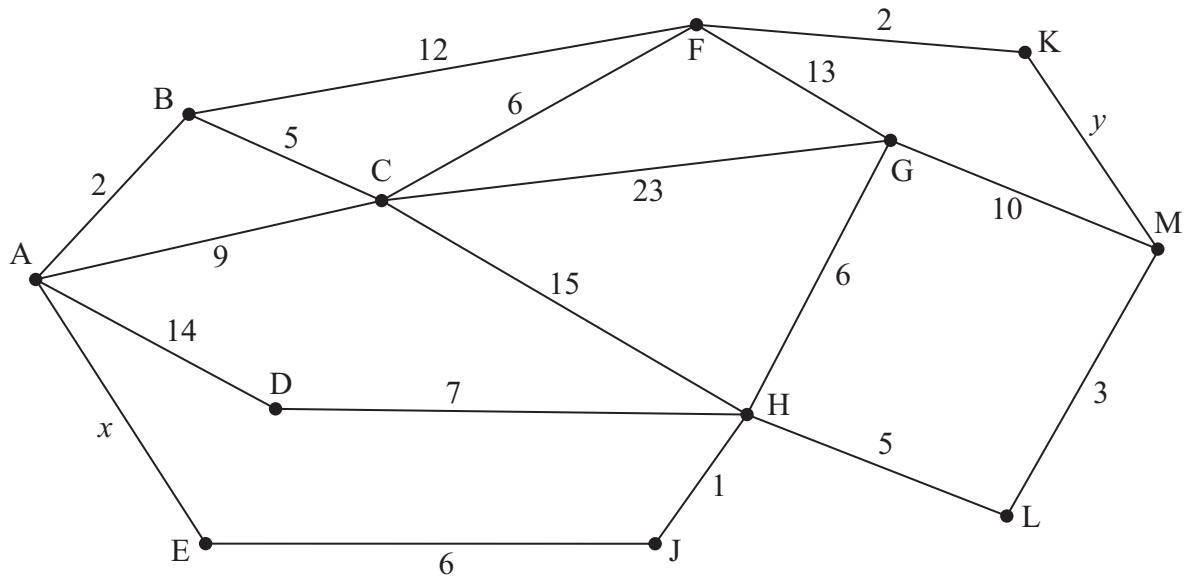


Figure 3

[The total weight of the network is $139 + x + y$]

(a) Explain what is meant by the term “tree”.

(1)

Figure 3 represents a network of walkways in a warehouse.

The arcs represent the walkways and the nodes represent junctions between them.

The number on each arc represents the length, in metres, of the corresponding walkway.

The values x and y are unknown, however it is known that x and y are integers and that

$$9 < x < y < 14$$

(b) (i) Use Dijkstra’s algorithm to find the shortest route from A to M.

(ii) State an expression for the length of the shortest route from A to M.

(6)

The warehouse manager wants to check that all of the walkways are in good condition.

Their inspection route starts at B and finishes at C.

The inspection route must traverse each walkway at least once and be as short as possible.

(c) State the arcs that are traversed twice.

(1)

(d) State the number of times that H appears in the inspection route.

(1)

The warehouse manager finds that the total length of the inspection route is 172 metres.

(e) Determine the value of x and the value of y

(2)

(Total for Question 3 is 11 marks)



4.

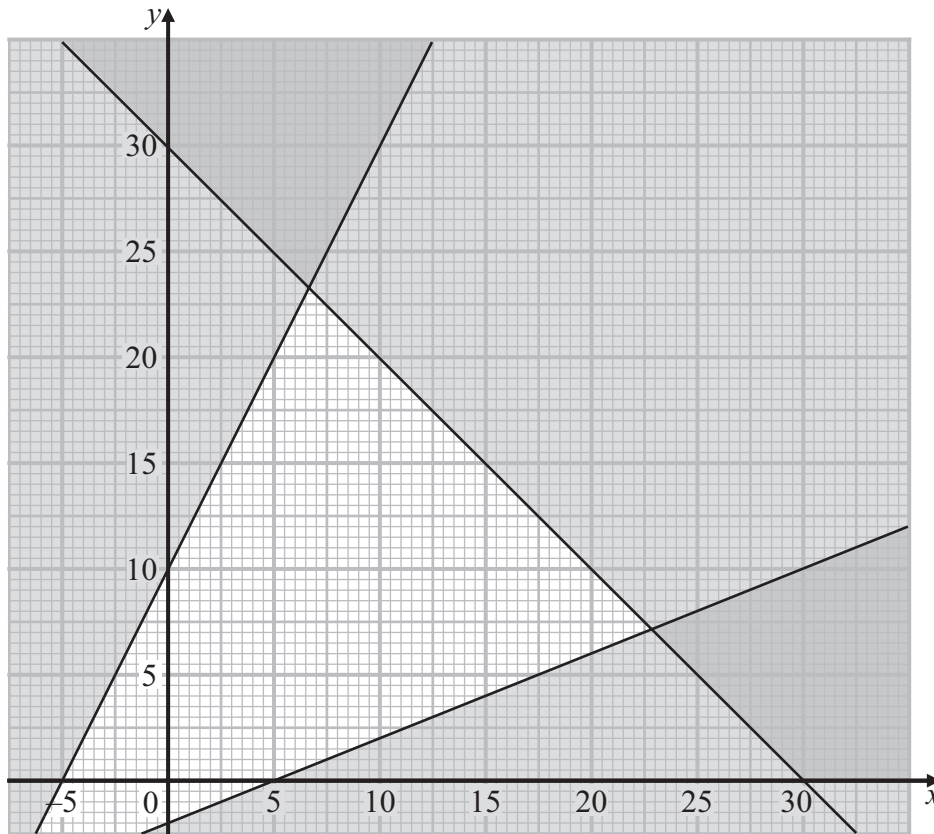


Figure 4

Figure 4 shows three of the six constraints for a linear programming problem in x and y . The unshaded region and its boundaries satisfy these three constraints.

(a) State these three constraints as simplified inequalities with integer coefficients.

(3)

The variables x and y represent the number of orange fish and the number of blue fish, respectively, that are to be kept in an aquarium.

The number of fish in the aquarium is subject to these three further constraints

- there must be at least one blue fish
- the orange fish must not outnumber the blue fish by more than ten
- there must be no more than five blue fish for every orange fish

(b) Write each of these three constraints as a simplified inequality with integer coefficients.

(2)

- (c) Represent these three constraints by adding lines and shading to Diagram 1 in the answer book, labelling the feasible region, R (3)

The total value (in pounds) of the fish in the aquarium is given by the objective function

$$\text{Maximise } P = 3x + 5y$$

- (d) (i) Use the objective line method to determine the optimal point of the feasible region, giving its coordinates as exact fractions.
- (ii) Hence find the maximum total value of the fish in the aquarium, stating the optimal number of orange fish and the optimal number of blue fish. (4)

(Total for Question 4 is 12 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS



Pearson Edexcel Level 3 GCE

Friday 19 May 2023

Afternoon

Paper
reference

8FM0/27

Further Mathematics

Advanced Subsidiary

Further Mathematics options

27: Decision Mathematics 1

(Part of options D, F, H and K)

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
D1 Answer Book (enclosed)

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the answer book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.
- Do not return the question paper with the D1 Answer Book.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P72813A

©2023 Pearson Education Ltd.
N:1/1/




Pearson

Answer ALL questions. Write your answers in the answer book provided.

1.

67 59 46 71 40 48 53 63 45 54 56

The list of eleven numbers shown above is to be sorted into descending order.

Carry out a quick sort to produce the sorted list. You should show the result of each pass and identify the pivots clearly.

(Total for Question 1 is 4 marks)



2.

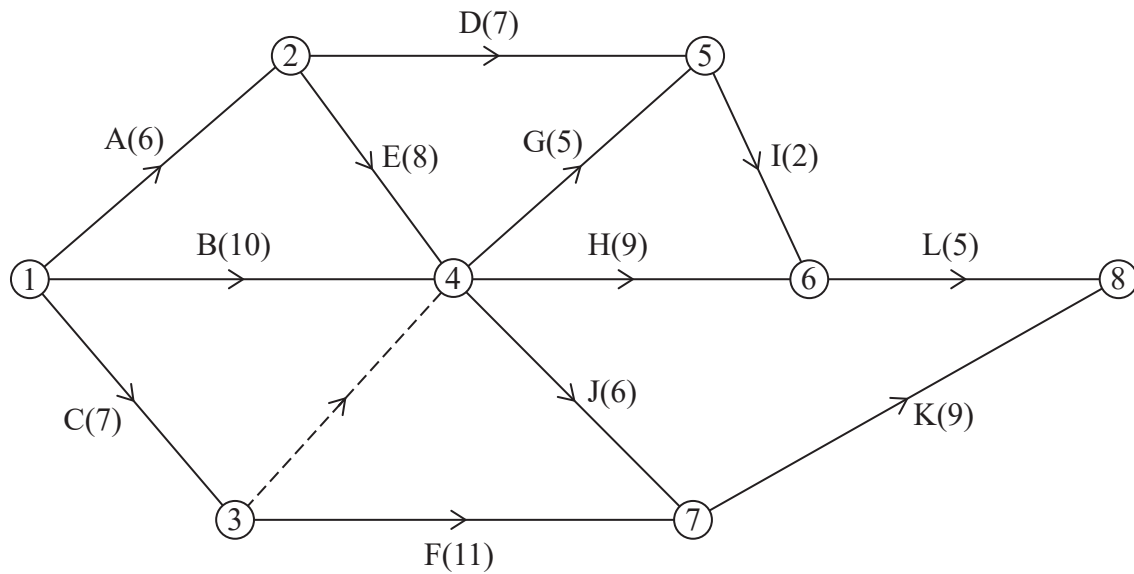


Figure 1

A project is modelled by the activity network shown in Figure 1. The activities are represented by the arcs. The number in brackets on each arc gives the time required, in hours, to complete the corresponding activity. The numbers in circles are the event numbers. Each activity requires one worker, and the project is to be completed in the shortest possible time.

- (a) Explain the significance of the dummy activity from event 3 to event 4 (1)
- (b) Complete Diagram 1 in the answer book to show the early event times and the late event times. (3)
- (c) State the critical activities. (1)
- (d) Calculate a lower bound for the number of workers needed to complete the project in the minimum time. You must show your working. (1)
- (e) Draw a Gantt chart for this project on Grid 1 in the answer book. (4)

(Total for Question 2 is 10 marks)

3.

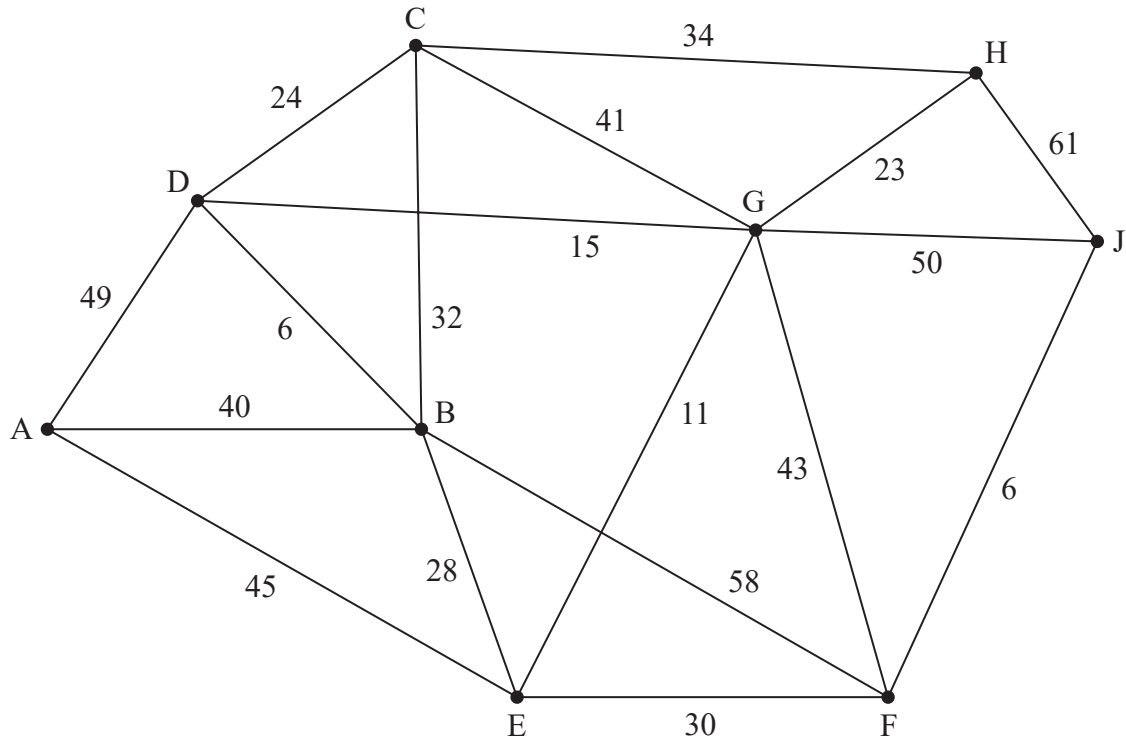


Figure 2

Figure 2 represents a network of train tracks. The number on each edge represents the length, in kilometres, of the corresponding track.

Dyfan wishes to travel from A to J via C. Dyfan wishes to minimise the distance they travel.

Given that Dijkstra's algorithm is to be applied only once to find Dyfan's route,

- (a) explain why the algorithm should begin at C. (1)
- (b) Use Dijkstra's algorithm to find the shortest route from A to J via C. State this route and its length. (6)
- (c) Use Prim's algorithm, starting at C, to find a minimum spanning tree for the network. You must clearly state the order in which you select the edges of your tree. (3)
- (d) State the total length, in km, of the minimum spanning tree. (1)

(Total for Question 3 is 11 marks)

4.

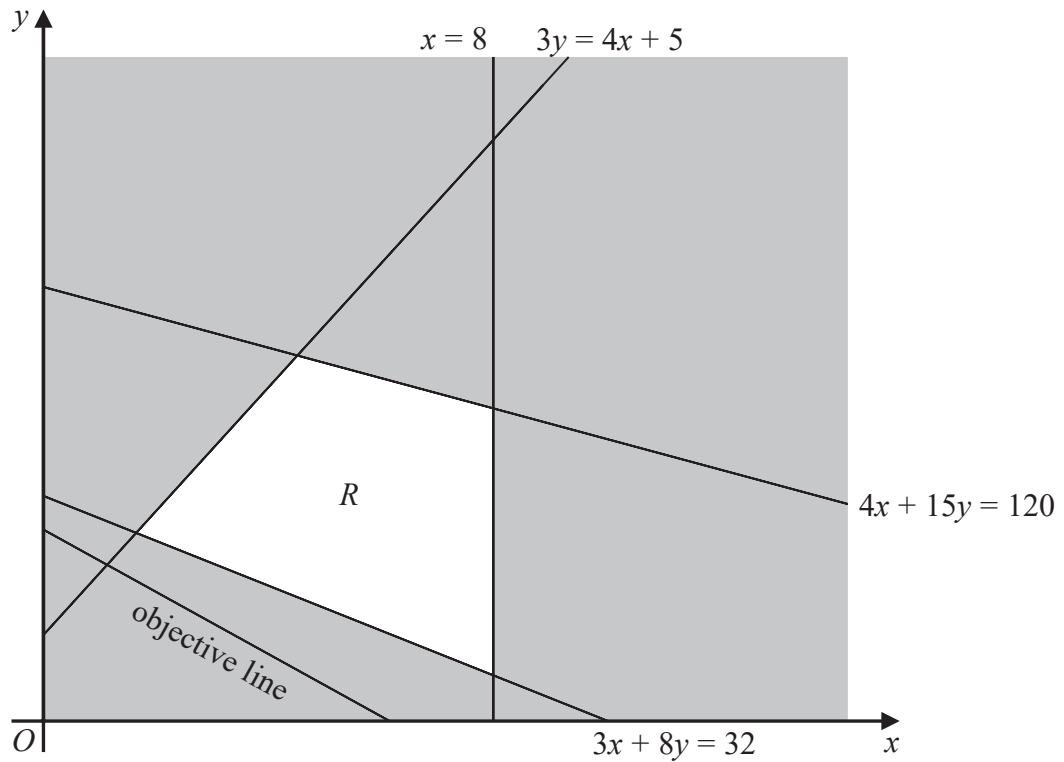


Figure 3

Figure 3 shows the constraints of a linear programming problem in x and y . The unshaded area, including its boundaries, forms the feasible region, R . An objective line has been drawn and labelled on the graph.

(a) State the inequalities that define the feasible region.

(2)

The maximum value of the objective function is $\frac{160}{3}$

The minimum value of the objective function is $\frac{883}{41}$

(b) Determine the objective function, showing your working clearly.

(5)

(Total for Question 4 is 7 marks)

5.

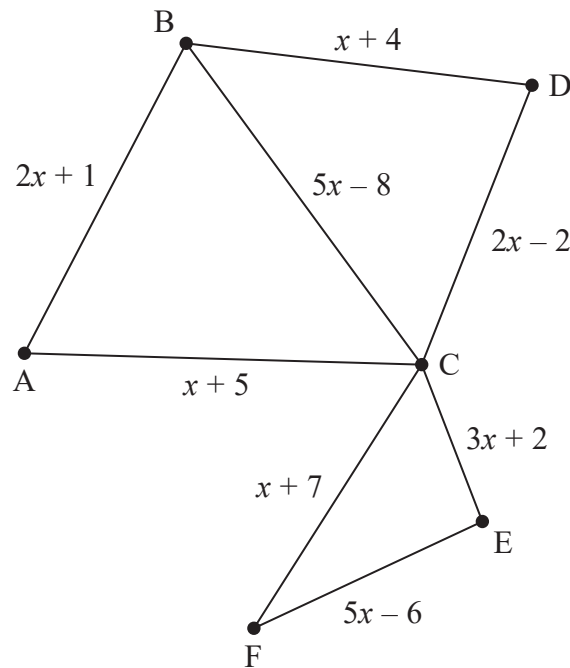


Figure 4

[The weight of the network is $20x + 3$]

Figure 4 shows a graph G that contains 8 arcs and 6 vertices.

- State the minimum number of arcs that would need to be added to make G into an Eulerian graph. (1)
- Explain whether or not the route $A - C - F - E - C - D - B$ is an example of a path on G . (1)

Figure 4 represents a network of 8 roads in a city. The expression on each arc gives the time, in minutes, to travel along the corresponding road.

You are given that $x > 1.6$

A route is required that

- starts and finishes at the same vertex
- traverses each road at least once
- minimises the total time taken

The route inspection algorithm is applied to the network in Figure 4 and the time taken for the route is found to be at most 189 minutes.

Given that the inspection route contains two roads that need to be traversed twice,

- determine the range of possible values of x , making your reasoning clear. (6)

(Total for Question 5 is 8 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS



Further Mathematics

Advanced Subsidiary

Further Mathematics options

27: Decision Mathematics 1

(Part of options D, F, H and K)

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
D1 Answer Book (enclosed)

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the answer book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.
- Do not return the question paper with the D1 Answer Book.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►



Answer ALL questions. Write your answers in the answer book provided.

1. 55 44 34 59 28 37 41 52 33 42 47

The list of eleven numbers shown above is to be sorted into ascending order.

- (a) Carry out a quick sort to produce the sorted list. You should show the result of each pass and identify your pivots clearly.

(4)

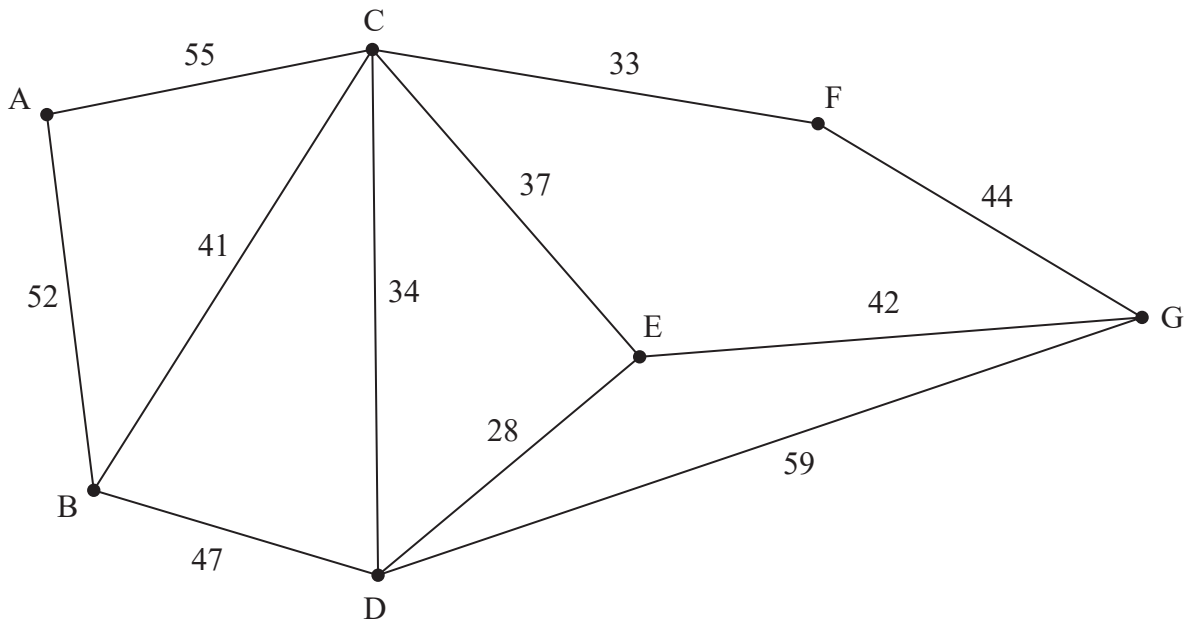


Figure 1

- (b) Use Kruskal's algorithm to find the minimum spanning tree for the network in Figure 1. You should list the arcs in the order in which you consider them. For each arc, state whether or not you are adding it to your minimum spanning tree.

(3)

- (c) (i) Draw the minimum spanning tree on Diagram 1 in the answer book.

- (ii) State the total weight of the tree.

(2)

(Total for Question 1 is 9 marks)

2.

Activity	Immediately preceding activities
A	–
B	–
C	–
D	–
E	A
F	A, B, C
G	C
H	C
I	E
J	E, F, G
K	D, H

(a) Draw the activity network described in the precedence table above, using activity on arc. Your activity network must contain the minimum number of dummies only. (5)

(b) Explain why it is necessary to draw a dummy from the end of activity A. (1)

Every activity shown in the precedence table has the same duration.

(c) State which activity cannot be critical, justifying your answer. (2)

(Total for Question 2 is 8 marks)

3.

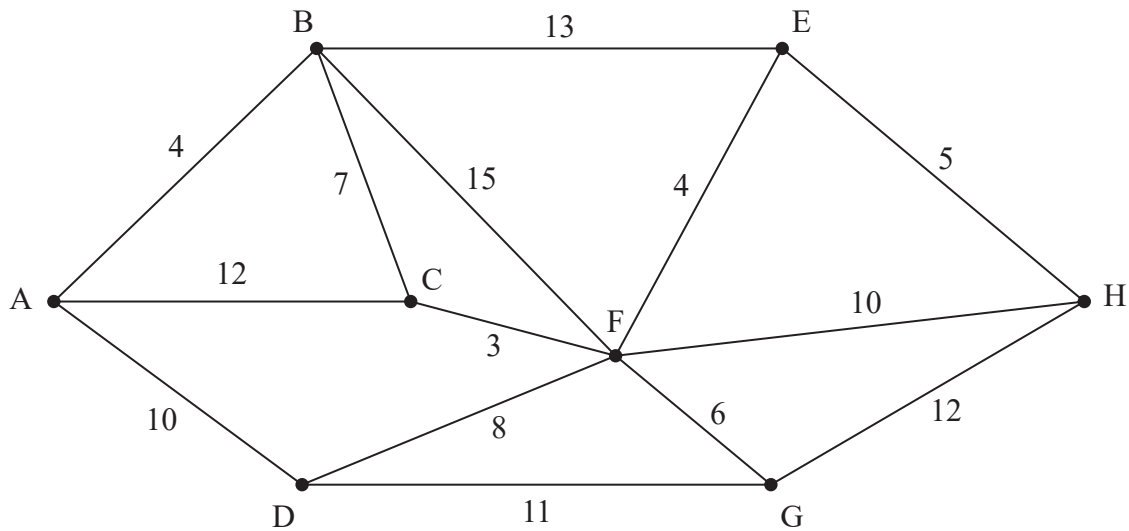


Figure 2

[The total weight of the network is 120]

(a) Explain what is meant by the term “path”. (2)

(b) State, with a reason, whether the network in Figure 2 is Eulerian, semi-Eulerian or neither. (1)

Figure 2 represents a network of cycle tracks between eight villages, A, B, C, D, E, F, G and H. The number on each arc represents the length, in km, of the corresponding track. Samira lives in village A, and wishes to visit her friend, Daisy, who lives in village H.

(c) Use Dijkstra’s algorithm to find the shortest path that Samira can take. (5)

An extra cycle track of length 9km is to be added to the network. It will either go directly between C and D or directly between E and G.

Daisy plans to cycle along every track in the new network, starting and finishing at H.

Given that the addition of either track CD or track EG will not affect the final values obtained in (c),

(d) use a suitable algorithm to find out which of the two possible extra tracks will give Daisy the shortest route, making your method and working clear. You must

- state which tracks Daisy will repeat in her route
- state the total length of her route

(6)

(Total for Question 3 is 14 marks)

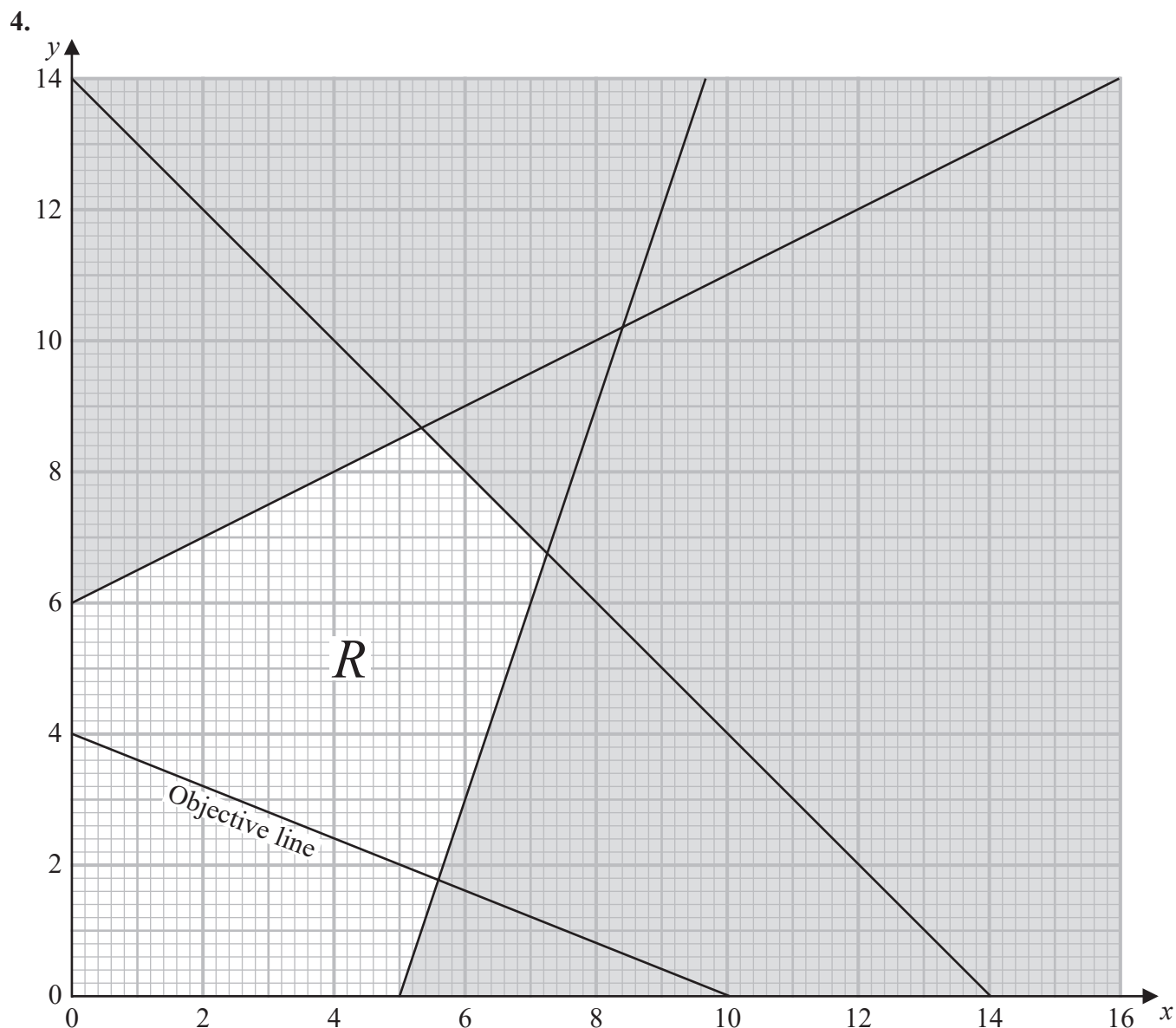


Figure 3

Figure 3 shows the constraints of a maximisation linear programming problem in x and y , where $x \geq 0$ and $y \geq 0$. The unshaded area, including its boundaries, forms the feasible region, R . An objective line has been drawn and labelled on the graph.

(a) List the constraints as simplified inequalities with integer coefficients.

(3)

The optimal value of the objective function is 216

(b) (i) Calculate the exact coordinates of the optimal vertex.

(ii) Hence derive the objective function.

(5)

Given that x represents the number of small flower pots and y represents the number of large flower pots supplied to a customer,

(c) deduce the optimal solution to the problem.

(1)

(Total for Question 4 is 9 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS

END

Further Mathematics

Advanced Subsidiary

Further Mathematics options

27: Decision Mathematics 1

(Part of options D, F, H and K)

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
D1 Answer Book (enclosed)

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the Answer Book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.
- Do not return the question paper with the D1 Answer Book.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

P66794A

©2021 Pearson Education Ltd.

1/1/1/1/



Write your answers in the answer book provided.

1. 3.5 6.3 2.9 5.4 3.1 2.8 3.7 1.7 4.1 3.3 2.2

The numbers listed above are to be sorted into descending order.

- (a) (i) Perform **one** pass of a bubble sort, starting at the left-hand end of the list. You must write down the list that results at the end of this first pass.
- (ii) Write down the number of comparisons and the number of swaps performed during this first pass.
- (3)**

After a second pass using this bubble sort, the updated list is

6.3 5.4 3.5 3.1 3.7 2.9 4.1 3.3 2.8 2.2 1.7

- (b) Use a quick sort on this updated list to obtain the fully sorted list. You should show the result of each pass and identify your pivots clearly.
- (3)**
- (c) Apply the first-fit decreasing bin packing algorithm to the fully sorted list to pack the numbers into bins of size 11.5
- (3)**
- (d) Determine whether your answer to part (c) uses the minimum number of bins. You must justify your answer.
- (2)**

(Total for Question 1 is 11 marks)

2.

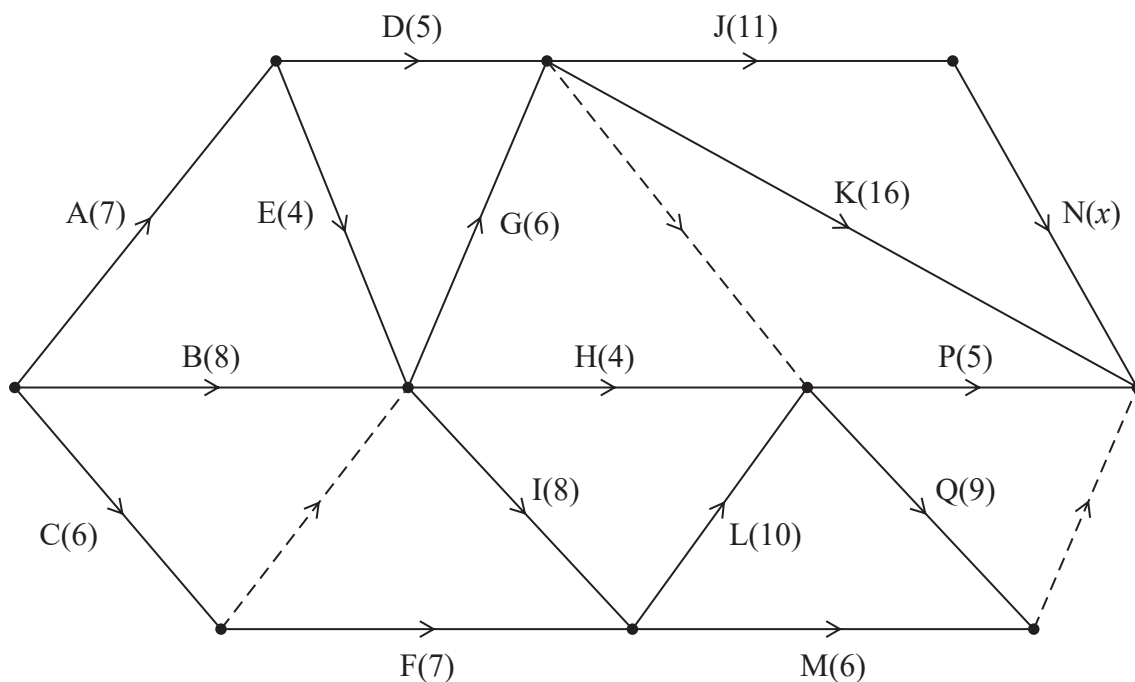


Figure 1

A project is modelled by the activity network shown in Figure 1. The activities are represented by the arcs. The number in brackets on each arc gives the time, in hours, to complete the corresponding activity. The exact duration, x , of activity N is unknown, but it is given that $5 < x < 10$

Each activity requires one worker. The project is to be completed in the shortest possible time.

- (a) Complete the precedence table in the answer book. (2)
 - (b) Complete Diagram 1 in the answer book to show the early event times and the late event times. (4)
 - (c) List the critical activities. (1)
- It is given that activity J can be delayed by up to 4 hours without affecting the shortest possible completion time of the project.
- (d) Determine the value of x . You must make the numbers used in your calculation clear. (1)
 - (e) Draw a cascade chart for this project on Grid 1 in the answer book. (4)

(Total for Question 2 is 12 marks)

3. Donald plans to bake and sell cakes. The three types of cake that he can bake are brownies, flapjacks and muffins.

Donald decides to bake 48 brownies and muffins in total.

Donald decides to bake at least 5 brownies for every 3 flapjacks.

At most 40% of the cakes will be muffins.

Donald has enough ingredients to bake 60 brownies or 45 flapjacks or 35 muffins.

Donald plans to sell each brownie for £1.50, each flapjack for £1 and each muffin for £1.25
He wants to maximise the total income from selling the cakes.

Let x represent the number of brownies, let y represent the number of flapjacks and let z represent the number of muffins that Donald will bake.

Formulate this as a linear programming problem in x and y only, stating the objective function and listing the constraints as simplified inequalities with integer coefficients.

You should **not** attempt to solve the problem.

(Total for Question 3 is 9 marks)

4.

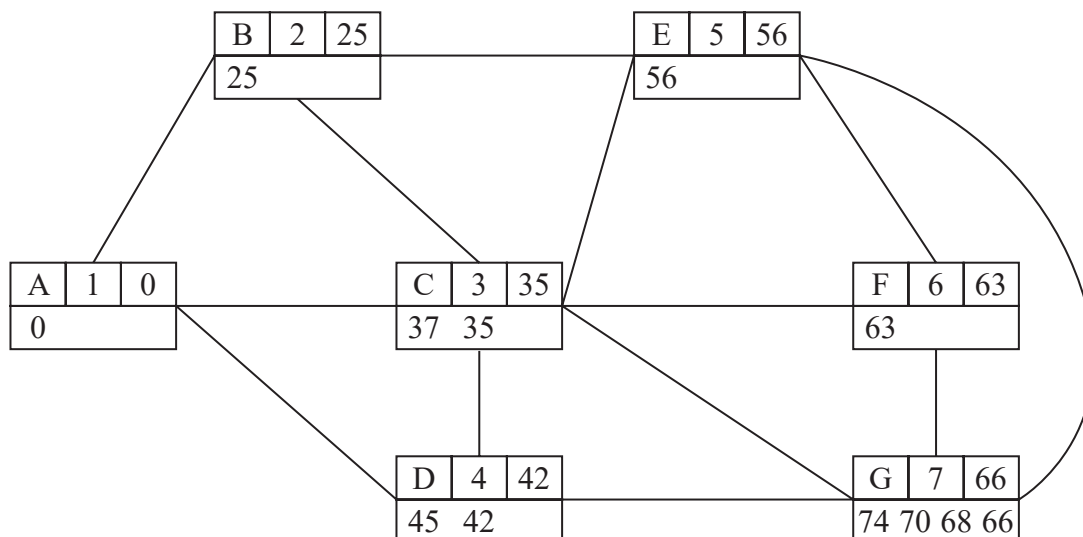


Figure 2

Dijkstra's algorithm has been applied to the network in Figure 2.

A working value has only been replaced at a node if the new working value is smaller.

- (a) State the length of the shortest path from A to G. (1)
- (b) Complete the table in the answer book giving the weight of each arc listed.
(Note that arc CE and arc EF are not in the table.) (3)
- (c) State the shortest path from A to G. (1)

It is now given that

- when Prim's algorithm, starting from A, is applied to the network, the order in which the arcs are added to the tree is AB, BC, CD, CE, EF and FG
- the weight of the corresponding minimum spanning tree is 80
- the shortest path from A to F via E has weight 67

- (d) Determine the weight of arc CE and the weight of arc EF, making your reasoning clear. (3)

(Total for Question 4 is 8 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS

END

Pearson Edexcel Level 3 GCE

Thursday 08 October 2020

Afternoon

Paper Reference **8FM0/27**

Further Mathematics

Advanced Subsidiary

Further Mathematics options

27: Decision Mathematics 1

(Part of options D, F, H and K)

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
D1 Answer Book (enclosed)

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the Answer Book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.
- Do not return the question paper with the D1 Answer Book.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P62676A

©2020 Pearson Education Ltd.

1/1/1/1/



P 6 2 6 7 6 A


Pearson

1. 3.7 2.5 5.4 1.9 2.7 3.2 3.1 2.7 4.2 2.0

- (a) Use the first-fit bin packing algorithm to determine how the numbers listed above can be packed into bins of size 8.5

(3)

The first-fit bin packing algorithm is to be used to pack n numbers into bins. The number of comparisons is used to measure the order of the first-fit bin packing algorithm.

- (b) By considering the worst case, determine the order of the first-fit bin packing algorithm in terms of n . You must make your method and working clear.

(3)

(Total for Question 1 is 6 marks)

2.

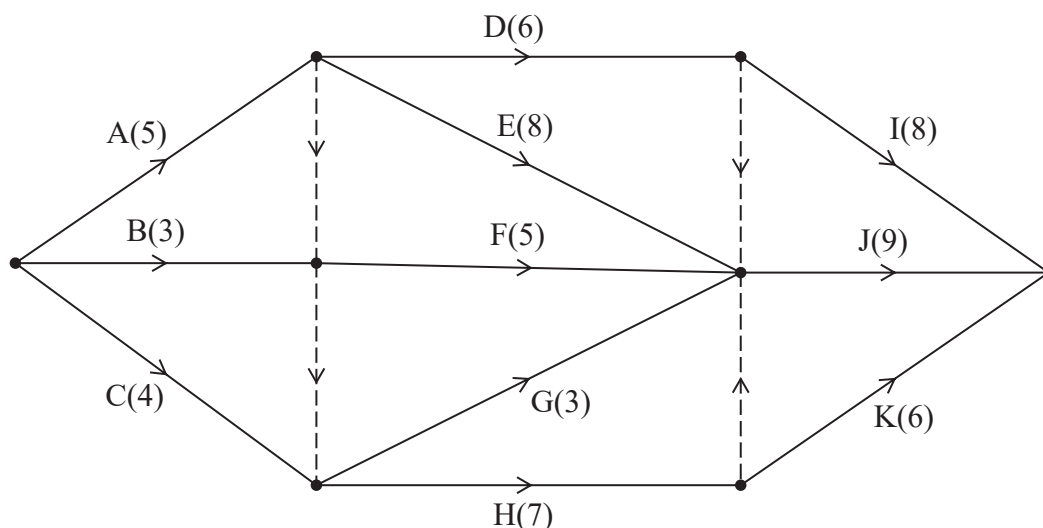


Figure 1

A project is modelled by the activity network shown in Figure 1. The activities are represented by the arcs. The number in brackets on each arc gives the time, in hours, to complete the corresponding activity. Each activity requires one worker. The project is to be completed in the shortest possible time.

- (a) Complete the precedence table in the answer book. (2)
- (b) Complete Diagram 1 in the answer book to show the early event times and the late event times. (3)
- (c) (i) State the minimum project completion time.
- (ii) List the critical activities. (2)
- (d) Calculate the maximum number of hours by which activity H could be delayed without affecting the shortest possible completion time of the project. You must make the numbers used in your calculation clear. (1)
- (e) Calculate a lower bound for the number of workers needed to complete the project in the minimum time. You must show your working. (2)
- (f) Draw a cascade chart for this project on Grid 1 in the answer book. (3)
- (g) Using the answer to (f), explain why it is not possible to complete the project in the shortest possible time using the number of workers found in (e). (1)

(Total for Question 2 is 14 marks)

3.

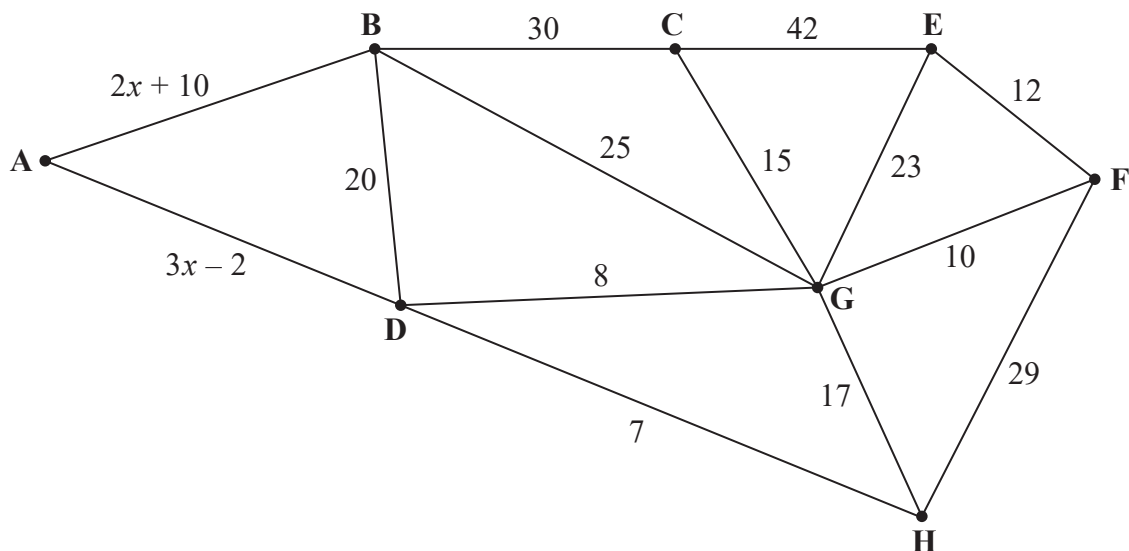


Figure 2

[The weight of the network is $5x + 246$]

- (a) Explain why it is not possible to draw a graph with an odd number of vertices of odd valency.

(2)

Figure 2 represents a network of 14 roads in a town. The expression on each arc gives the time, in minutes, to travel along the corresponding road.

Prim's algorithm, starting at A, is applied to the network. The order in which the arcs are selected is AD, DH, DG, FG, EF, CG, BD. It is given that the order in which the arcs are selected is unique.

- (b) Using this information, find the smallest possible range of values for x , showing your working clearly.

(3)

A route that minimises the total time taken to traverse each road at least once is required. The route must start and finish at the same vertex.

Given that the time taken to traverse this route is 318 minutes,

- (c) use an appropriate algorithm to determine the value of x , showing your working clearly.

(6)

(Total for Question 3 is 11 marks)

4.

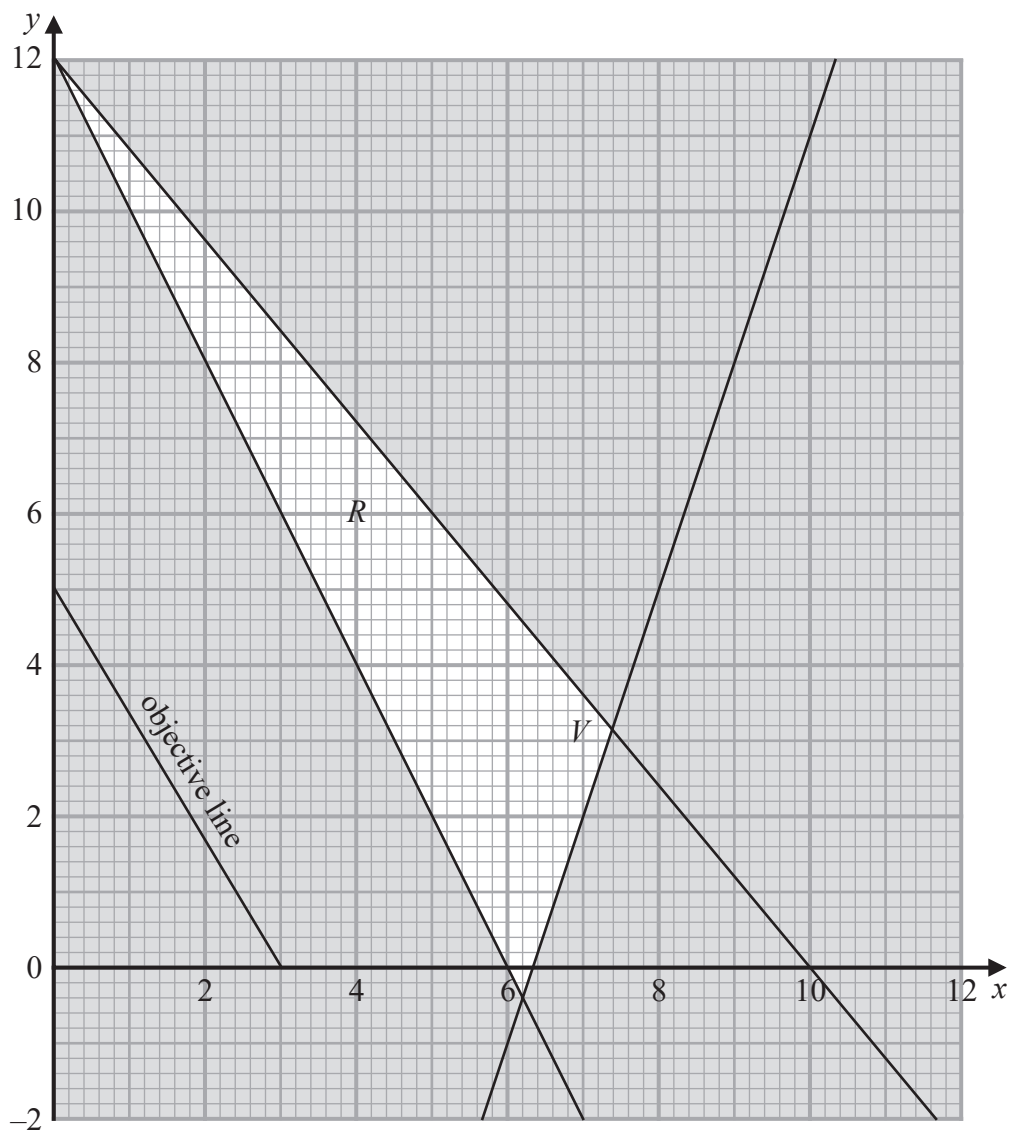


Figure 3

Figure 3 shows the constraints of a linear programming problem in x and y , where R is the feasible region. Figure 3 also shows an objective line for the problem and the optimal vertex, which is labelled as V .

The value of the objective at V is 556

Express the linear programming problem in algebraic form. List the constraints as simplified inequalities with integer coefficients and determine the objective.

(9)

(Total for Question 4 is 9 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS

END

Pearson Edexcel Level 3 GCE

Thursday 16 May 2019

Afternoon

Paper Reference **8FM0-27**

Further Mathematics

Advanced Subsidiary

Further Mathematics options

27: Decision Mathematics 1

(Part of options D, F, H and K)

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
D1 Answer Book (enclosed)

**Candidates may use any calculator allowed by Pearson regulations.
Calculators must not have the facility for symbolic algebra manipulation,
differentiation and integration, or have retrievable mathematical
formulae stored in them.**

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the Answer Book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 5 questions.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P61639A

©2019 Pearson Education Ltd.

1/1/1/1/1/1/



P 6 1 6 3 9 A


Pearson

1. (a) Draw the graph K_5 (1)
- (b) (i) In the context of graph theory explain what is meant by ‘semi-Eulerian’.
- (ii) Draw two semi-Eulerian subgraphs of K_5 , each having five vertices but with a different number of edges. (3)
- (c) Explain why a graph with exactly five vertices with vertex orders 1, 2, 2, 3 and 4 cannot be a tree. (2)

(Total for Question 1 is 6 marks)

2. The following algorithm produces a numerical approximation for the integral

$$I = \int_A^B x^4 \, dx$$

Step 1	Start
Step 2	Input the values of A, B and N
Step 3	Let $H = (B - A) / N$
Step 4	Let $C = H / 2$
Step 5	Let $D = 0$
Step 6	Let $D = D + A^4 + B^4$
Step 7	Let $E = A$
Step 8	Let $E = E + H$
Step 9	If $E = B$ go to Step 12
Step 10	Let $D = D + 2 \times E^4$
Step 11	Go to Step 8
Step 12	Let $F = C \times D$
Step 13	Output F
Step 14	Stop

For the case when $A = 1$, $B = 3$ and $N = 4$,

- (a) (i) complete the table in the answer book to show the results obtained at each step of the algorithm.
- (ii) State the final output. (4)
- (b) Calculate, to 3 significant figures, the percentage error between the exact value of I and the value obtained from using the approximation to I in this case. (3)

(Total for Question 2 is 7 marks)

3.

Activity	Immediately preceding activities
A	-
B	-
C	A
D	A
E	A
F	B, C
G	B, C
H	D
I	D, E, F, G
J	D, E, F, G
K	G

- (a) Draw the activity network described in the precedence table above, using activity on arc.
Your activity network must contain the minimum number of dummies.

(5)

Every activity shown in the precedence table has the same duration.

- (b) Explain why activity B cannot be critical.

(1)

- (c) State which other activities are not critical.

(1)

(Total for Question 3 is 7 marks)

4.

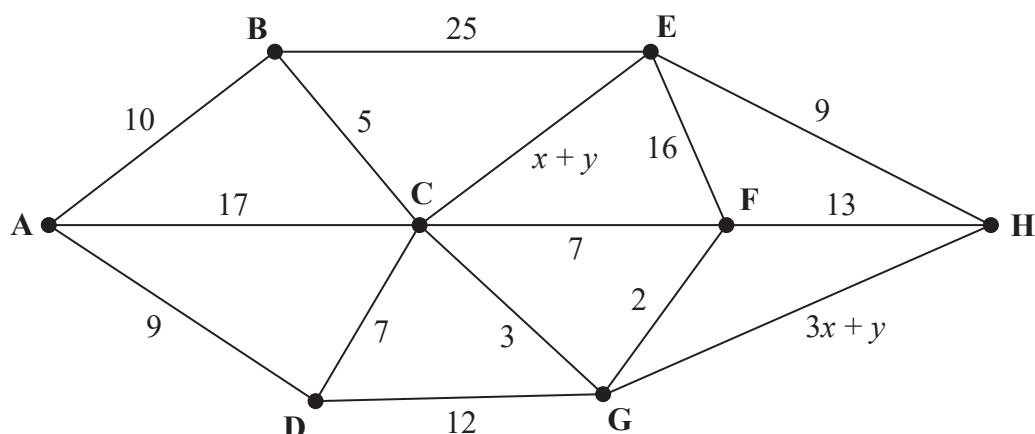


Figure 1

[The total weight of the network is $135 + 4x + 2y$]

The weights on the arcs in Figure 1 represent distances. The weights on the arcs CE and GH are given in terms of x and y , where x and y are positive constants and $7 < x + y < 20$

There are three paths from A to H that have the same minimum length.

(a) Use Dijkstra's algorithm to find x and y .

(7)

An inspection route starting at A and finishing at H is found. The route traverses each arc at least once and is of minimum length.

(b) State the arcs that are traversed twice.

(1)

(c) State the number of times that vertex C appears in the inspection route.

(1)

(d) Determine the length of the inspection route.

(1)

(Total for Question 4 is 10 marks)

5. Ben is a wedding planner. He needs to order flowers for the weddings that are taking place next month. The three types of flower he needs to order are roses, hydrangeas and peonies.

Based on his experience, Ben forms the following constraints on the number of each type of flower he will need to order.

- At least three-fifths of all the flowers must be roses.
- For every 2 hydrangeas there must be at most 3 peonies.
- The total number of flowers must be exactly 1000

The cost of each rose is £1, the cost of each hydrangea is £5 and the cost of each peony is £4

Ben wants to minimise the cost of the flowers.

Let x represent the number of roses, let y represent the number of hydrangeas and let z represent the number of peonies that he will order.

- (a) Formulate this as a linear programming problem in x and y only, stating the objective function and listing the constraints as simplified inequalities with integer coefficients. (7)

Ben decides to order the minimum number of roses that satisfy his constraints.

- (b) (i) Calculate the number of each type of flower that he will order to minimise the cost of the flowers.
- (ii) Calculate the corresponding total cost of this order. (3)

(Total for Question 5 is 10 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS

END

Pearson Edexcel Level 3 GCE

Further Mathematics

Advanced Subsidiary
Further Mathematics options
27: Decision Mathematics 1
(Part of options D, F, H and K)

Thursday 17 May 2018 – Afternoon

Paper Reference

8FM0-27

You must have:

Mathematical Formulae and Statistical Tables, calculator,
D1 Answer Book (enclosed)

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the Answer Book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P60208A

©2018 Pearson Education Ltd.

1/1/1/




Pearson

Answer ALL questions. Write your answers in the Answer Book provided.

1.

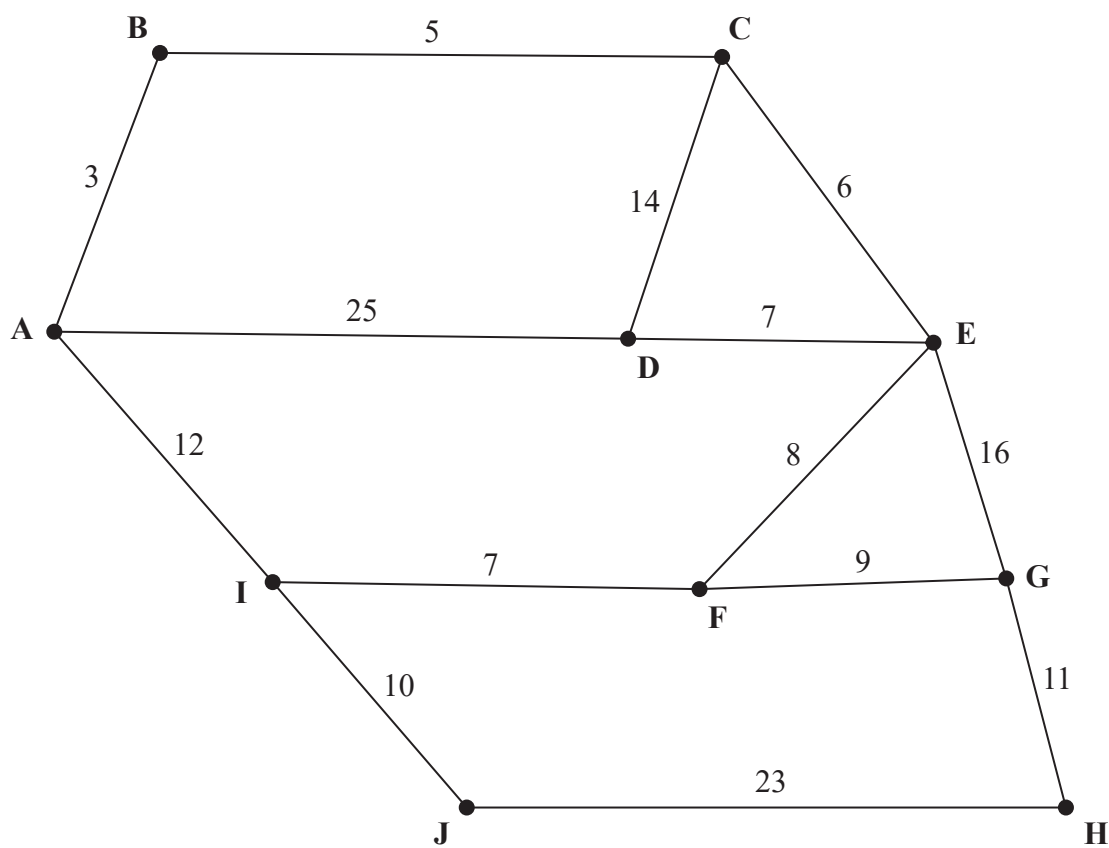


Figure 1

Figure 1 represents a network of roads.

The number on each arc represents the time taken, in minutes, to drive along the corresponding road.

(a) (i) Use Dijkstra's algorithm to find the shortest time needed to travel from A to H.

(ii) State the quickest route.

(6)

For a network with n vertices, Dijkstra's algorithm has order n^2

(b) If it takes 1.5 seconds to run the algorithm when $n = 250$, calculate approximately how long it will take, in seconds, to run the algorithm when $n = 9500$. You should make your method and working clear.

(2)

(c) Explain why your answer to part (b) is only an approximation.

(1)

(Total for Question 1 is 9 marks)

2. A simply connected graph is a connected graph in which any two vertices are directly connected by at most one arc and no vertex is directly connected to itself.

(a) Given that a simply connected graph has exactly four vertices,

(i) write down the minimum number of arcs it can have,

(ii) write down the maximum number of arcs it can have.

(2)

(b) (i) Draw a simply connected graph that has exactly four vertices and exactly five arcs.

(ii) State, with justification, whether your graph is Eulerian, semi-Eulerian or neither.

(3)

(c) By considering the orders of the vertices, explain why there is only one simply connected graph with exactly four vertices and exactly five arcs.

(5)

(Total for Question 2 is 10 marks)

3.

Activity	Time taken (days)	Immediately preceding activities
A	5	–
B	8	–
C	4	–
D	14	A
E	10	A
F	3	B, C, E
G	7	C
H	5	D, F, G
I	7	H
J	9	H

The table above shows the activities required for the completion of a building project. For each activity, the table shows the time it takes, in days, and the immediately preceding activities. Each activity requires one worker. The project is to be completed in the shortest possible time.

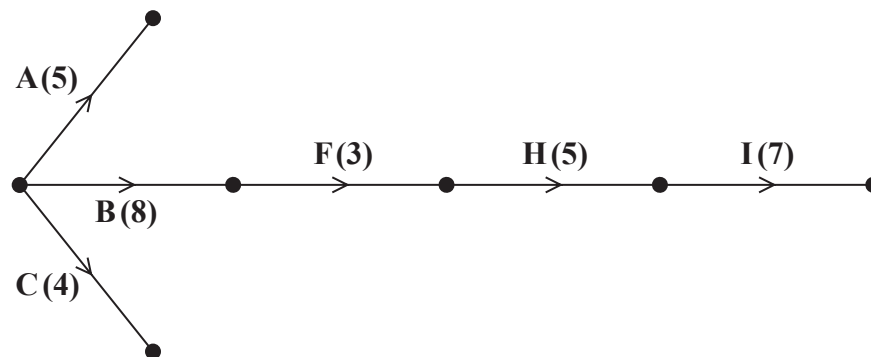


Figure 2

Figure 2 shows a partially completed activity network used to model the project. The activities are represented by the arcs and the number in brackets on each arc is the time taken, in days, to complete the corresponding activity.

- Add the missing activities and necessary dummies to Diagram 1 in the answer book. (3)
- Complete Diagram 1 in the answer book to show the early event times and the late event times. (3)
- State the critical activities. (1)

At the beginning of the project it is decided that activity G is no longer required.

- Explain what effect, if any, this will have on
 - the shortest completion time of the project if activity G is no longer required,
 - the timing of the remaining activities.
- (3)

(Total for Question 3 is 10 marks)

4. The manager of a factory is planning the production schedule for the next three weeks for a range of cabinets. The following constraints apply to the production schedule.
- The total number of cabinets produced in week 3 cannot be fewer than the total number produced in weeks 1 and 2
 - At most twice as many cabinets must be produced in week 3 as in week 2
 - The number of cabinets produced in weeks 2 and 3 must, in total, be at most 125

The production cost for each cabinet produced in weeks 1, 2 and 3 is £250, £275 and £200 respectively.

The factory manager decides to formulate a linear programming problem to find a production schedule that minimises the total cost of production.

The objective is to minimise $250x + 275y + 200z$

(a) Explain what the variables x , y and z represent. (1)

(b) Write down the constraints of the linear programming problem in terms of x , y and z . (2)

Due to demand, exactly 150 cabinets must be produced during these three weeks. This reduces the constraints to

$$\begin{aligned}x + y &\leq 75 \\x + 3y &\geq 150 \\x &\geq 25 \\y &\geq 0\end{aligned}$$

which are shown in Diagram 1 in the answer book.

Given that the manager does not want any cabinets left unfinished at the end of a week,

(c) (i) use a graphical approach to solve the linear programming problem and hence determine the production schedule which minimises the cost of production. You should make your method and working clear.

(ii) Find the minimum total cost of the production schedule. (8)

(Total for Question 4 is 11 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS

END

Write your name here

Surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

--	--	--	--	--

Candidate Number

--	--	--	--	--

Further Mathematics

Advanced Subsidiary

Further Mathematics options

Paper 2D: Section B Decision Mathematics 1

Sample Assessment Material for first teaching September 2017

Questions 6 – 10

Paper Reference

8FM0/2D

Do not return this document with the question paper.

Total Marks

S58541A

©2017 Pearson Education Ltd.

1/1/1/1/1/1/



S 5 8 5 4 1 A 0 1 0 6

Turn over ►


Pearson

SECTION B

Answer ALL questions. Write your answers in the answer book provided.

6.

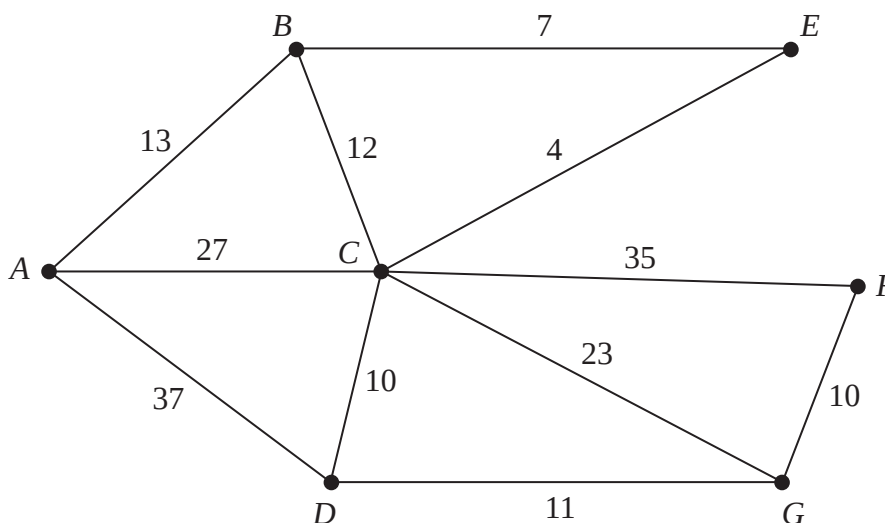


Figure 1

[The total weight of the network is 189]

Figure 1 represents a network of pipes in a building. The number on each arc is the length, in metres, of the corresponding pipe.

- (a) Use Dijkstra's algorithm to find the shortest path from A to F. State the path and its length. (5)

On a particular day, Gabriel needs to check each pipe. A route of minimum length, which traverses each pipe at least once and which starts and finishes at A, needs to be found.

- (b) Use an appropriate algorithm to find the pipes that will need to be traversed twice. You must make your method and working clear. (4)

- (c) State the minimum length of Gabriel's route. (1)

A new pipe, BG, is added to the network. A route of minimum length that traverses each pipe, including BG, needs to be found. The route must start and finish at A.

Gabriel works out that the addition of the new pipe increases the length of the route by twice the length of BG.

- (d) Calculate the length of BG. You must show your working. (2)

(Total for Question 6 is 12 marks)

7.

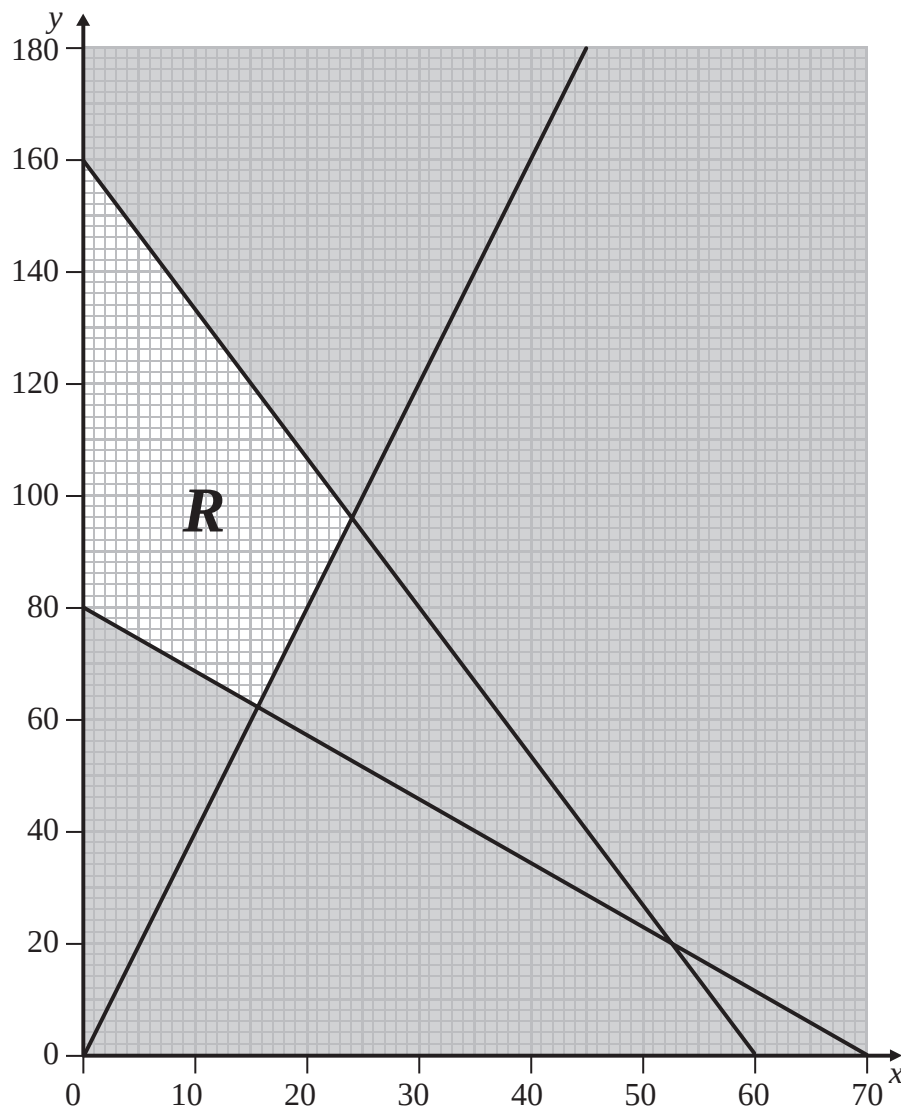


Figure 2

A teacher buys pens and pencils. The number of pens, x , and the number of pencils, y , that he buys can be represented by a linear programming problem as shown in Figure 2, which models the following constraints:

$$8x + 3y \leq 480$$

$$8x + 7y \geq 560$$

$$y \geq 4x$$

$$x, y \geq 0$$

The total cost, in pence, of buying the pens and pencils is given by

$$C = 12x + 15y$$

Determine the number of pens and the number of pencils which should be bought in order to minimise the total cost. You should make your method and working clear.

(Total for Question 7 is 7 marks)

8.

Activity	Time taken (days)	Immediately preceding activities
A	5	-
B	7	-
C	3	-
D	4	A, B
E	4	D
F	2	B
G	4	B
H	5	C, G
I	10	C, G

The table above shows the activities required for the completion of a building project. For each activity, the table shows the time taken in days to complete the activity and the immediately preceding activities. Each activity requires one worker. The project is to be completed in the shortest possible time.

- (a) Draw the activity network described in the table, using activity on arc. Your activity network must contain the minimum number of dummies only. (3)
- (b) (i) Show that the project can be completed in 21 days, showing your working. (4)
- (ii) Identify the critical activities.

(Total for Question 8 is 7 marks)

9. (a) Explain why it is not possible to draw a graph with exactly 5 nodes with orders 1, 3, 4, 4 and 5 (1)

A connected graph has exactly 5 nodes and contains 18 arcs. The orders of the 5 nodes are $2^{2x} - 1$, 2^x , $x + 1$, $2^{x+1} - 3$ and $11 - x$.

- (b) (i) Calculate x .
(ii) State whether the graph is Eulerian, semi-Eulerian or neither. You must justify your answer. (6)

- (c) Draw a graph which satisfies all of the following conditions:

- The graph has exactly 5 nodes.
- The nodes have orders 2, 2, 4, 4 and 4
- The graph is not Eulerian.

(2)

(Total for Question 9 is 9 marks)

10. Jonathan makes two types of information pack for an event, *Standard* and *Value*.

Each *Standard* pack contains 25 posters and 500 flyers.

Each *Value* pack contains 15 posters and 800 flyers.

He must use at least 150 000 flyers.

Between 35% and 65% of the packs must be *Standard* packs.

Posters cost 20p each and flyers cost 4p each.

Jonathan wishes to minimise his costs.

Let x and y represent the number of *Standard* packs and *Value* packs produced respectively.

Formulate this as a linear programming problem, stating the objective and listing the constraints as simplified inequalities with integer coefficients.

You should not attempt to solve the problem.

(Total for Question 10 is 5 marks)

TOTAL FOR SECTION B IS 40 MARKS

Pearson Edexcel Level 3 GCE

Specimen Paper

Paper Reference **8FM0/27**

Further Mathematics

Advanced Subsidiary

27: Decision Mathematics 1

You must have:

Mathematical Formulae and Statistical Tables, calculator,
D2 Answer Book (enclosed)

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the Answer Book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

S60746A

©2018 Pearson Education Ltd.

1/




Pearson

Answer ALL questions. Write your answers in the answer book provided.

1. The following algorithm determines the total number of comparisons required when Prim's algorithm is applied to a complete network on n vertices, where $n > 2$

Step 1 Input the value of n

Step 2 Let $A = 1$

Step 3 Let $B = n - 2$

Step 4 Let $C = B$

Step 5 Let $A = A + 1$

Step 6 Let $B = B - 1$

Step 7 Let $C = C + (A \times B) + (A - 1)$

Step 8 If $B > 0$ go to Step 5, otherwise go to Step 9

Step 9 Output C

Step 10 Stop

- (a) (i) For the case when $n = 6$, complete the table in the answer book to show the results obtained at each step when the algorithm is applied.

(ii) State the final output.

(4)

$$f(n) = \frac{1}{6}n^3 + kn + 1, \quad n \in \mathbb{N}, n > 2$$

The function $f(n)$, gives the same output as the algorithm.

- (b) Determine the value of k .

(2)

- (c) By considering $f(n)$, state what can be inferred about the efficiency of Prim's algorithm.

(1)

(Total for Question 1 is 7 marks)

2.

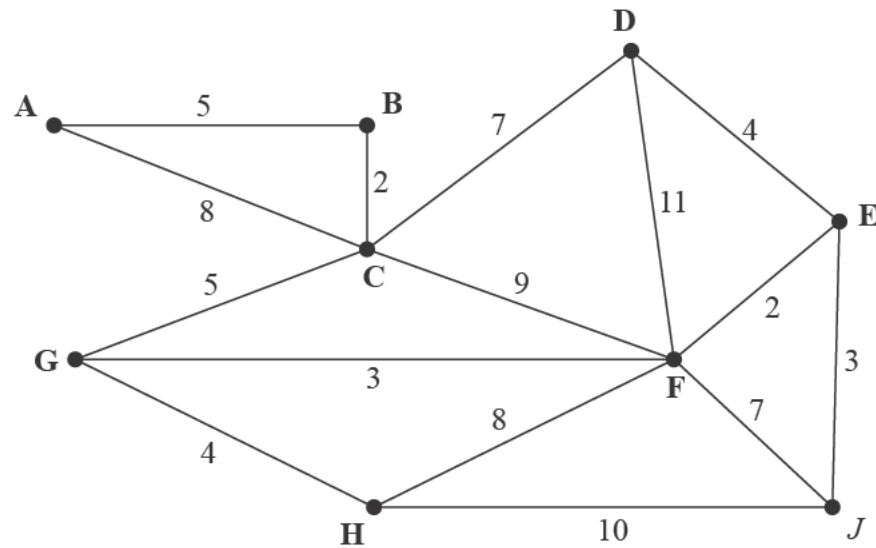


Figure 1

[The total weight of the network is 88]

The weights on the arcs in Figure 1 represent distances.

(a) (i) Use Dijkstra's algorithm to find the shortest path from A to J.

(ii) State the length of the shortest path from A to J.

(5)

An arc of weight 6 is added to the network between nodes C and H.

An inspection route of minimum length that traverses each arc at least once needs to be found. The inspection route must start and finish at vertex A.

(b) Use an appropriate algorithm to find the arcs that will need to be traversed twice. You must make your method and working clear.

(4)

(c) Determine the length of the inspection route.

(1)

(Total for Question 2 is 10 marks)

3.

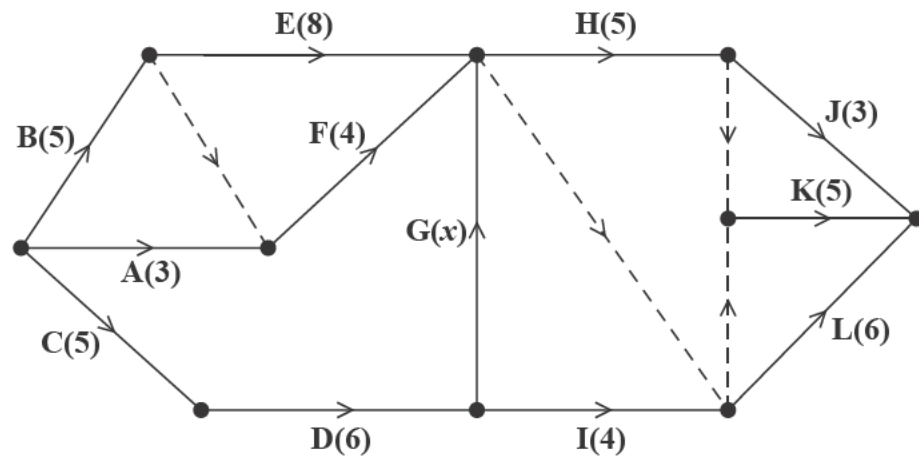


Figure 2

A project is modelled by the activity network shown in Figure 2. The activities are represented by the arcs. The number in brackets on each arc gives the time, in days, to complete the corresponding activity. Each activity requires exactly one worker. The project is to be completed in the shortest possible time.

Before the project begins, the duration of activity G, in days, is unknown and is given by x , where $x > 8$

- (a) Complete Diagram 1 in the answer book to show the early event times and the late event times, in terms of x where necessary. (4)

- (b) Identify the critical activities. (1)

The sum of the total float for activity A and the total float for activity E is equal to the minimum completion time of the project.

- (c) Making your reasoning clear, determine the minimum completion time for the project. (4)

(Total for Question 3 is 9 marks)

4. A baker makes lemon cakes and cherry cakes for a forthcoming fair.

The baker must make at least 360 cakes in total.

The cost of the ingredients for each lemon cake is 40p and the cost of the ingredients for each cherry cake is 20p. The baker has a total budget of £200 to spend on ingredients.

The baker must make at least twice as many cherry cakes as lemon cakes and he must make at least 100 lemon cakes.

The baker calculates that the time required to decorate each lemon cake is 1 minute and the time required to decorate each cherry cake is 3 minutes. The baker wants to minimise the total time needed to decorate all the required cakes.

- (a) Formulate this information as a linear programming problem. (6)
- (b) Use a graphical method to determine the number of each type of cake the baker should make in order to minimise the total decoration time. You should make your method and working clear. (7)
- (c) Using your solution to part (b), determine how much of the baker's budget for ingredients will be unspent. (1)

(Total for Question 4 is 14 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS

END